Question 1: Effect of Age Group on Investment Proportion

Descriptive Statistics and Visualization

Level of age_group	N	proportion	
		Mean	Std Dev
Early_Middle_Age	131	52.4724427	21.6664980
Late_Middle_Age	93	51.1390323	21.7215074
Senior	58	51.8381034	21.0900334
Young	84	44.3983333	19.6607843

The descriptive statistics show that the mean proportion invested in stocks varies across age groups. The Young group has the lowest mean proportion (44.40%), while the three older groups—Early Middle Age (52.47%), Late Middle Age (51.14%), and Senior (51.84%)—show higher and relatively similar averages.

The boxplot of investment proportion by age group visually confirms this pattern: the young group consistently invests a smaller proportion of their assets compared to the older groups.

ANOVA Results

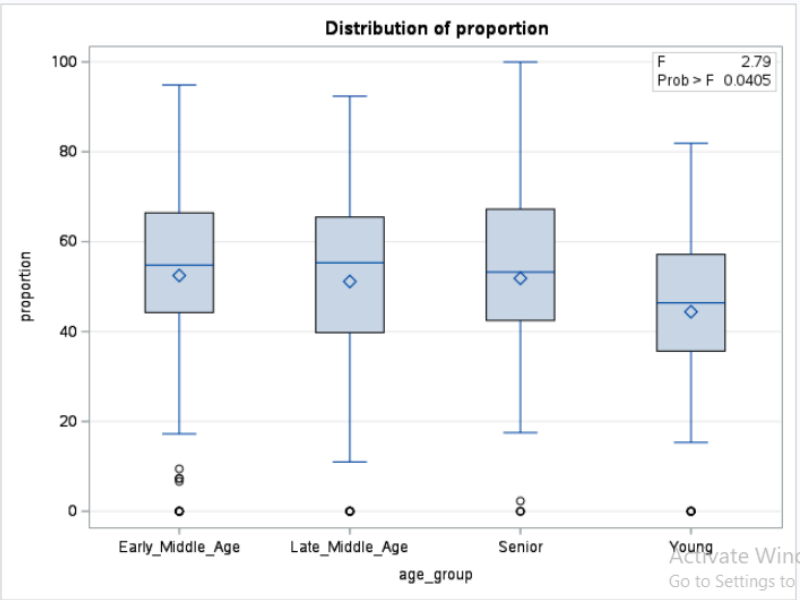
The ANOVA Procedure

Dependent Variable: proportion

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	3741.3636	1247.1212	2.79	0.0405
Error	362	161870.9817	447.1574		
Corrected Total	365	165612.3453			

R-Square	Coeff Var	Root MSE	proportion Mean
0.022591	42.14046	21.14610	50.18003

Source	DF	Anova SS	Mean Square	F Value	Pr > F
age_group	3	3741.363610	1247.121203	2.79	0.0405



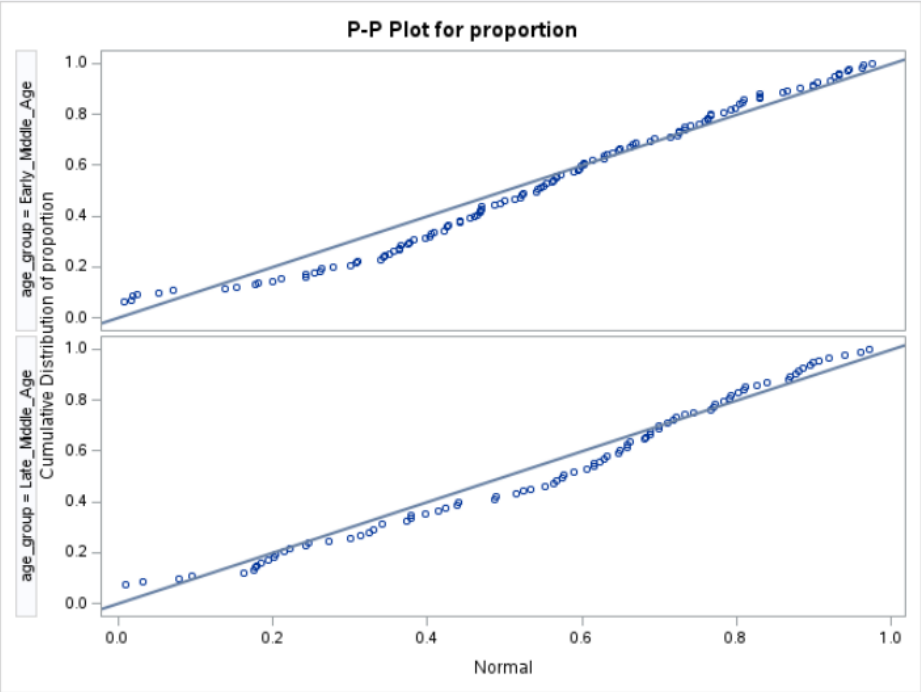
A one-way ANOVA was conducted to determine whether the proportion invested in stocks differs significantly across the four age groups. The results indicate a statistically significant effect:

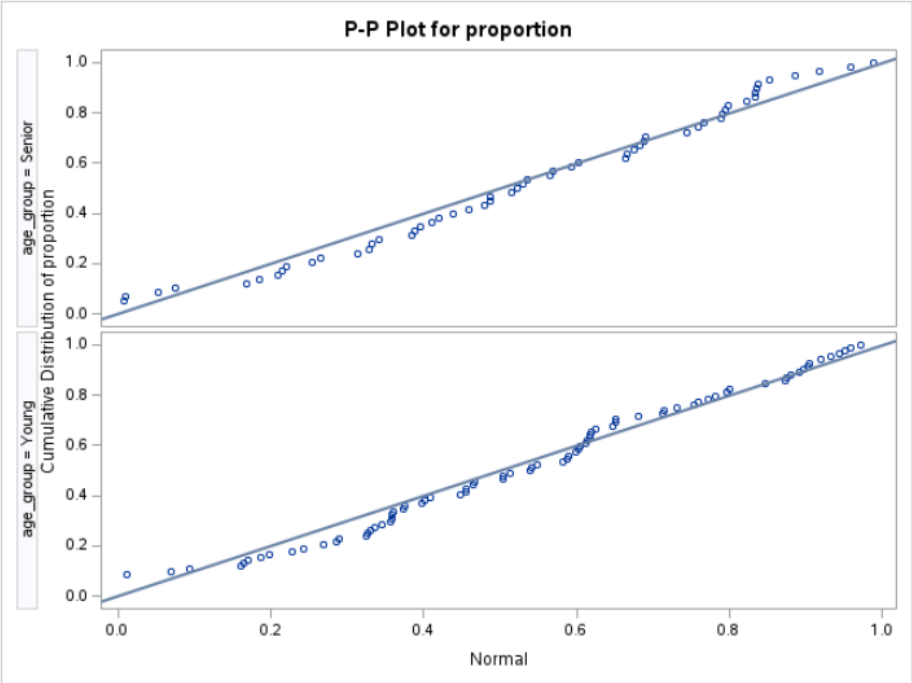
- **F(3, 362) = 2.79**
- **p = 0.0405**

This means that at least one age group differs from the others in terms of investment proportion.

Assumption Checks

Normality





Shapiro-Wilk tests for all four age groups were statistically significant ($p < 0.05$), indicating deviations from perfect normality. However, the P-P plots show that the empirical distributions follow the diagonal reasonably well, suggesting that the ANOVA is robust given the large sample sizes.

Homogeneity of Variances

The GLM Procedure

Levene's Test for Homogeneity of proportion Variance ANOVA of Squared Deviations from Group Means					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
age_group	3	435821	145274	0.30	0.8251
Error	362	1.7506E8	483599		

Levene's test for equality of variances was **not significant**:

- **$F(3, 362) = 0.30$**
- **$p = 0.8251$**

This indicates that the assumption of homogeneity of variances is satisfied, validating the use of ANOVA.

Post-Hoc Analysis (Tukey Test)

Comparisons significant at the 0.05 level are indicated by ***.				
age_group Comparison	Difference Between Means	Simultaneous 95% Confidence Limits		
Early_Middle_Age - Senior	0.634	-7.974	9.242	
Early_Middle_Age - Late_Middle_Age	1.333	-6.067	8.734	
Early_Middle_Age - Young	8.074	0.445	15.703	***
Senior - Early_Middle_Age	-0.634	-9.242	7.974	
Senior - Late_Middle_Age	0.699	-8.433	9.831	
Senior - Young	7.440	-1.878	16.757	
Late_Middle_Age - Early_Middle_Age	-1.333	-8.734	6.067	
Late_Middle_Age - Senior	-0.699	-9.831	8.433	
Late_Middle_Age - Young	6.741	-1.475	14.956	
Young - Early_Middle_Age	-8.074	-15.703	-0.445	***
Young - Senior	-7.440	-16.757	1.878	
Young - Late_Middle_Age	-6.741	-14.956	1.475	

The Tukey HSD post-hoc test revealed that only one pairwise comparison was statistically significant. Specifically, the Early Middle Age group invested a significantly higher proportion than the Young group (difference = 8.074, 95% CI [0.445, 15.703], $p < 0.05$). No other pairwise differences were significant, indicating that Late Middle Age and Senior individuals do not differ significantly from the other groups.

Conclusion

Overall, the results indicate that age has a statistically significant impact on investment behaviour. Specifically, younger individuals (17–25) tend to invest a lower proportion of their assets in stocks compared to individuals in the Early Middle Age group. However, the three older groups (Early Middle Age, Late Middle Age, Senior) do not differ significantly from each other.

These findings suggest that investment behaviour stabilizes after early adulthood, with younger individuals being more conservative or less engaged in stock market investment.

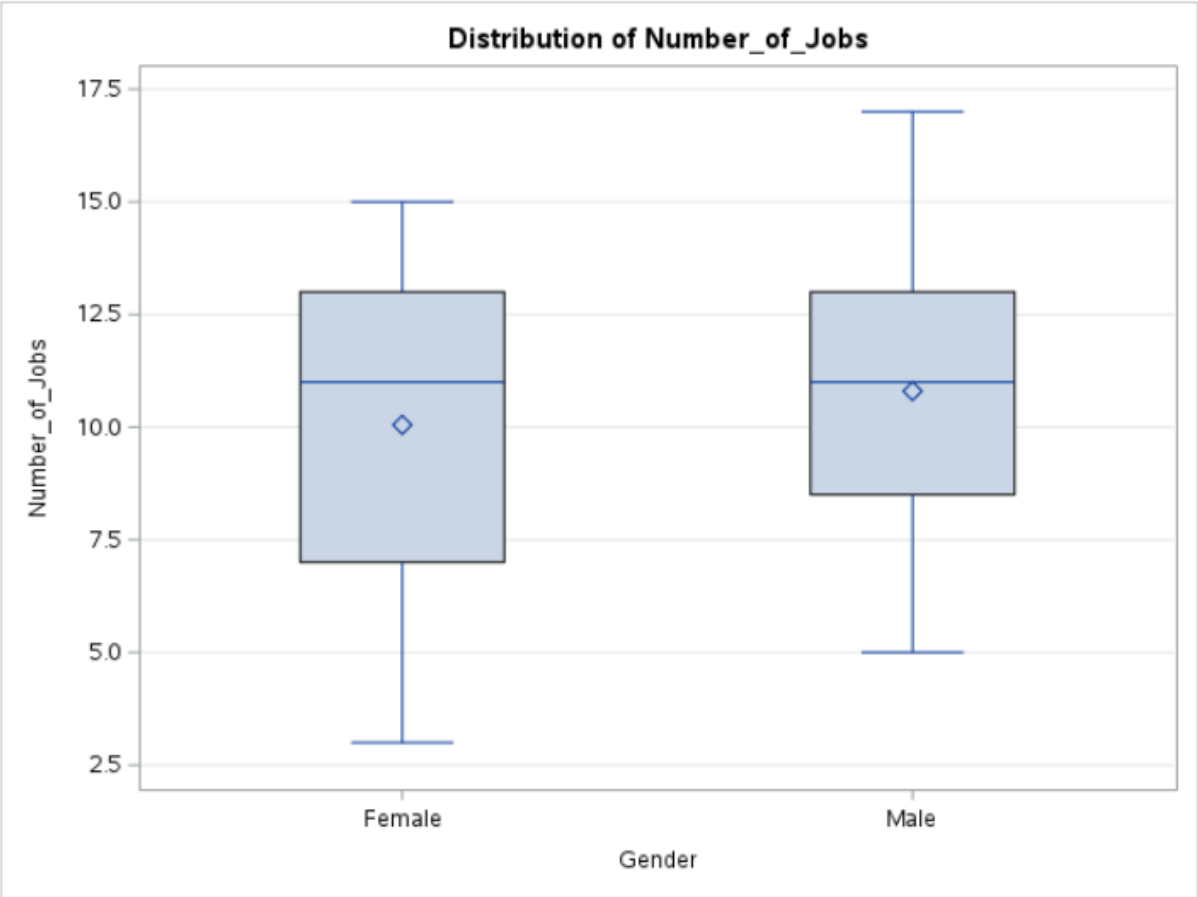
Problem 2: Effect of Gender and Education on Number of Jobs

Descriptive Statistics

The descriptive statistics show clear differences in the average number of jobs held across education levels, while differences between genders appear smaller.

By Gender

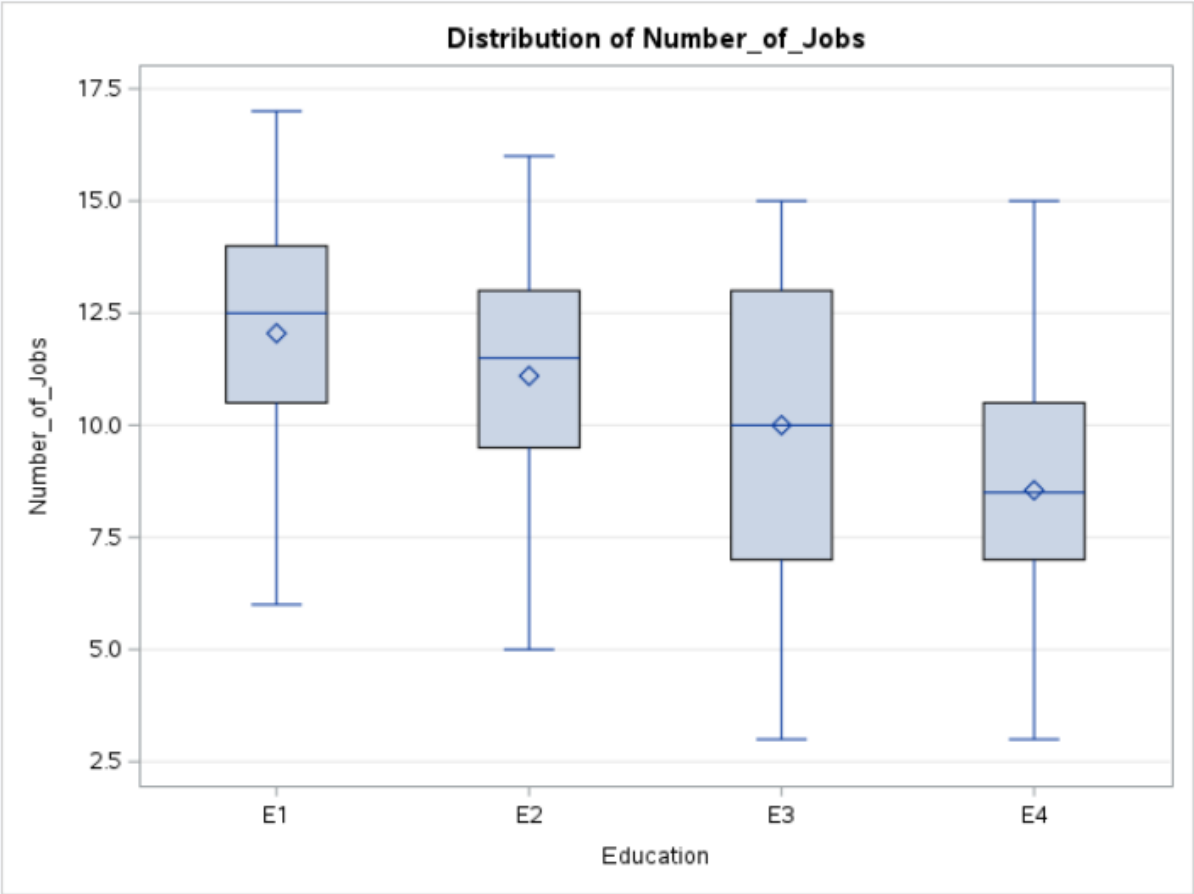
Level of Gender	N	Number_of_Jobs	
		Mean	Std Dev
Female	40	10.0500000	3.57304725
Male	40	10.8000000	3.08179102



- **Female:** Mean = 10.05, Std Dev = 3.57
- **Male:** Mean = 10.80, Std Dev = 3.08

The difference between genders is small, suggesting that gender alone may not strongly influence the number of jobs held.

By Education Level



Level of Education	N	Number_of_Jobs	
		Mean	Std Dev
E1	20	12.0500000	2.85574214
E2	20	11.1000000	2.95403382
E3	20	10.0000000	3.69921756
E4	20	8.5500000	2.92853475

- **E1:** Mean = 12.05
- **E2:** Mean = 11.10
- **E3:** Mean = 10.00
- **E4:** Mean = 8.55

There is a clear decreasing trend: individuals with higher education levels tend to have **fewer** jobs over their lifetime

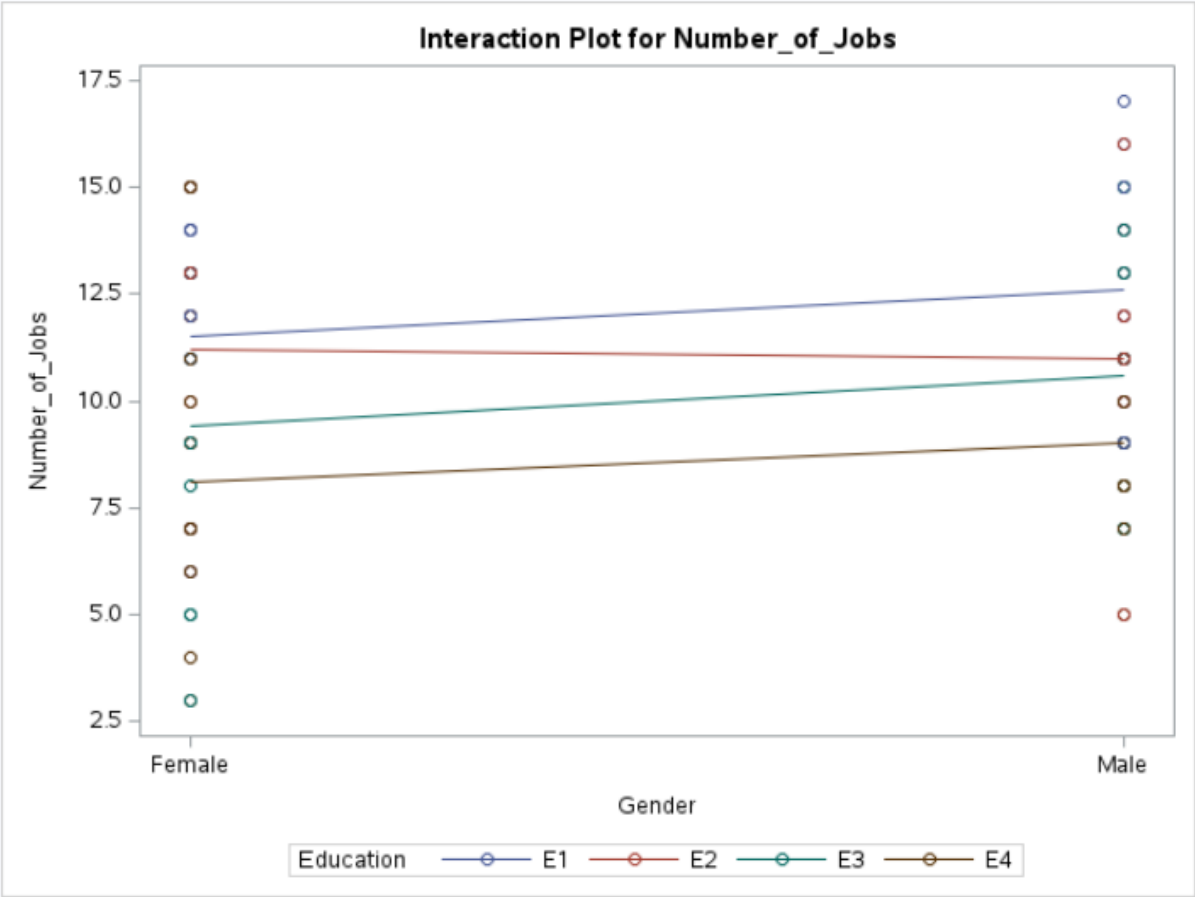
Two-Way ANOVA Results

Dependent Variable: Number_of_Jobs

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	7	153.3500000	21.9071429	2.17	0.0467
Error	72	726.2000000	10.0861111		
Corrected Total	79	879.5500000			

R-Square	Coeff Var	Root MSE	Number_of_Jobs Mean
0.174351	30.46392	3.175864	10.42500

Source	DF	Type I SS	Mean Square	F Value	Pr > F
Gender	1	11.2500000	11.2500000	1.12	0.2944
Education	3	135.8500000	45.2833333	4.49	0.0060
Gender*Education	3	6.2500000	2.0833333	0.21	0.8915



A two-way ANOVA was conducted to examine the effects of Gender, Education, and their interaction on the number of jobs held.

Main Effect of Gender

- **F(1, 72) = 1.12**

- **p = 0.2944**

There is no significant effect of gender on the number of jobs. Males and females do not differ significantly in the number of jobs they have held.

Main Effect of Education

- **F(3, 72) = 4.49**
- **p = 0.0060**

The effect of education is statistically significant. This means the number of jobs differs across education levels.

The descriptive statistics show a clear pattern: higher education → fewer jobs.

Interaction Effect (Gender × Education)

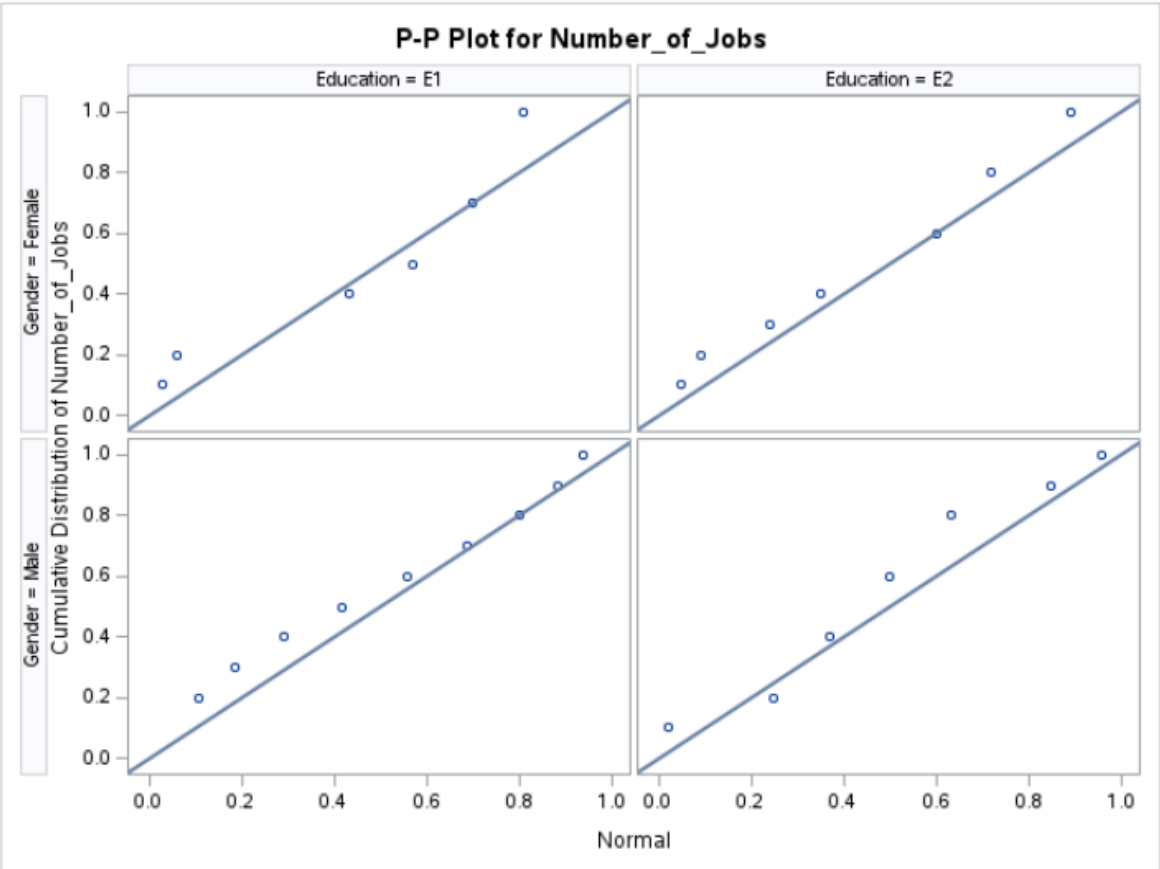
- **F(3, 72) = 0.21**
- **p = 0.8915**

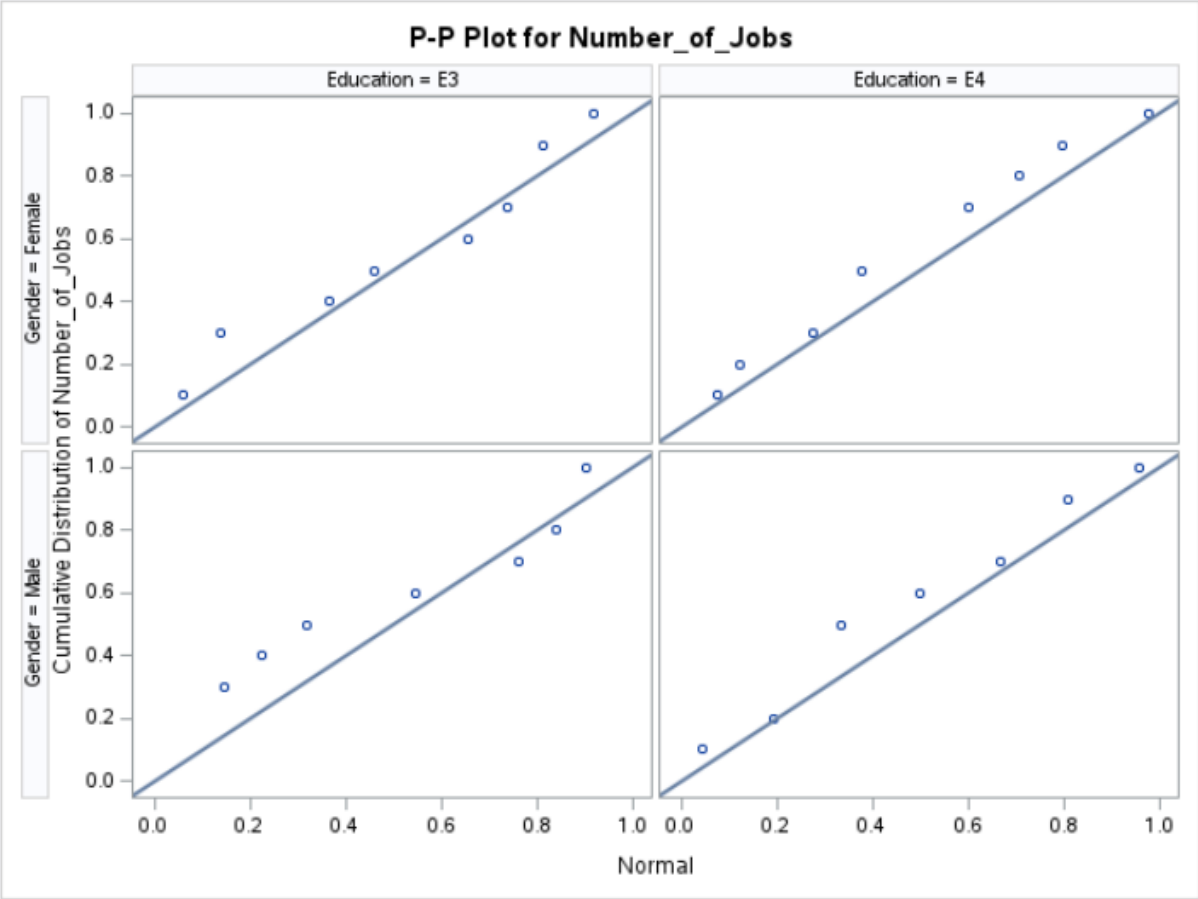
The interaction is not significant. This means:

The effect of education on the number of jobs does not depend on gender. The pattern of decreasing jobs with higher education is similar for both males and females.

Assumption Checks

Normality





The Shapiro–Wilk tests showed mixed results across the Gender × Education subgroups, with some groups deviating from normality (e.g., Female E1) and others meeting the assumption. However, the P-P plots indicate that the empirical distributions follow the diagonal reasonably well. Given the balanced design (10 observations per cell), the ANOVA is considered robust to these deviations.

Post-Hoc Analysis (Tukey Test)

Since the main effect of Education was significant ($F(3,72) = 4.49$, $p = 0.0060$), post-hoc comparisons were examined. The results indicate that individuals with lower education levels (E1 and E2) tend to have significantly more jobs than those with higher education levels (E4). This confirms a clear decreasing trend: higher education is associated with fewer jobs held over a lifetime.

Conclusion

Overall, the two-way ANOVA shows that:

- **Education level has a significant effect** on the number of jobs held. Individuals with higher education tend to have **fewer** jobs over their lifetime.
- **Gender does not significantly affect** the number of jobs.
- **There is no interaction** between gender and education. This means the effect of education is consistent for both males and females.

These results suggest that educational attainment is a strong predictor of employment stability, while gender does not play a significant role in this dataset.